

SupportLens AI

Final Evaluation Report — Hiver Support Agent Take-Home

Project goal. Build a small support agent that classifies an incoming customer message, uses historical support conversations as evidence for a reply, and decides whether to auto-handle or escalate the case.

81% useful historical evidence @5	41.33% Golden Set intent accuracy	20% AUTO-HANDLE / 80% ESCALATE
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1. What I Built

I focused the prototype on AppleSupport from the Customer Support on Twitter dataset. The pipeline first predicts a primary support intent, then retrieves five similar historical customer messages and examines the AppleSupport replies for repeated resolution actions. A simple deterministic drafter turns those supported actions into a response, while a conservative policy decides whether the case has enough evidence to auto-handle.

The main idea was to avoid generating an answer just because a model is confident. The response should have some historical support behind it. If that evidence is weak, the safer outcome is escalation.

2. Dataset and Evaluation Setup

The raw dataset contains about 2.8M tweets. After selecting AppleSupport replies and joining them to their customer messages, I obtained 106,260 usable customer → AppleSupport pairs. The raw dataset is kept outside the final repository because of its size and dataset-use considerations.

I created a locked 196-example Golden Set and labelled each example using the customer's primary problem. I also manually reviewed 100 retrieval queries (five retrieved cases each) and 100 historical replies for resolution-action extraction.

3. Pipeline

Customer message	→	Intent classifier	→	Historical retrieval
Historical replies	→	Resolution evidence	→	Grounded draft + decision

This is a prototype rather than a production support system. The design deliberately favors traceability and reproducibility over a more sophisticated but harder-to-explain architecture.

Approach

The implementation was kept lightweight so that the complete evaluation can run locally without a paid LLM API.

4. Intent Classification

The final taxonomy contains 12 intents: IOS_UPDATE, BATTERY_POWER, APP_PROBLEM, CONNECTIVITY, ACCOUNT_APPLE_ID, APP_STORE_PURCHASE, MUSIC_MEDIA, SCREEN_DISPLAY, SETTINGS_FEATURE, PERFORMANCE_STABILITY, HARDWARE_DEVICE, OTHER.

The raw dataset does not contain these intent labels, so I used high-confidence heuristic rules to bootstrap training labels. Rules use strong phrases and keywords, abstain on weak or ambiguous cases, and produced 13,098 high-confidence weak-labelled examples.

The classifier is TF-IDF with unigrams and bigrams followed by Logistic Regression with balanced class weights. The important evaluation choice was to keep the manually labelled Golden Set independent of this weak-label process.

5. Historical Retrieval

Historical customer messages are indexed with TF-IDF. For each new message, the system retrieves the top five similar customer cases and keeps the associated AppleSupport replies. I did not treat the highest cosine similarity as proof that the case is relevant.

Instead, the replies are passed through a small resolution-action extractor. Examples include UPDATE_SOFTWARE, CHECK_SETTINGS, CHECK_CONNECTION, CHECK_ACCOUNT, PROVIDE_TROUBLESHOOTING, PROVIDE_WORKAROUND, ASK_FOR_DETAILS, CONTACT_SUPPORT_DM and routing actions.

An action becomes stronger evidence when it appears in multiple retrieved cases. This was chosen because one historical reply can be noisy or very specific to its original customer.

6. Response Drafting

The final response is intentionally deterministic. It is not presented as an LLM-generated answer. Supported actions are converted into a small set of response templates. This makes the result easy to trace: an action in the draft should be explainable by retrieved historical evidence.

For example, if several historical battery cases consistently lead to a direct-message handoff and requesting device details, the draft uses those actions instead of inventing battery advice that was not observed in the evidence.

7. Escalation

The policy escalates when the predicted intent is OTHER, intent confidence is below 0.60, there is no supported action, or the strongest action is not observed in at least two retrieved cases. Otherwise it may auto-handle by producing the grounded first response.

AUTO-HANDLE therefore means automatic first-response generation, not a claim that the underlying customer problem has definitely been fixed.

Evaluation Results

The evaluation was designed to separate simple baselines, intent generalization, historical evidence quality, and the final agent behavior.

8. Baselines and Intent Results

System	Accuracy	Macro F1	Evaluation set
Majority class	11.73%	1.75%	196 Golden
Keyword baseline	50.51%	47.14%	196 Golden
TF-IDF + Logistic Regression	41.33%	41.68%	196 Golden
Weak-label validation	94.73%	92.73%	Weak-label holdout

The keyword baseline currently beats the learned classifier on the independent Golden Set. This is a useful negative result: the weak-label training process is not yet giving robust intent generalization.

The 94.73% validation number is included for transparency, but it is not treated as the headline result because both training and validation labels came from the same heuristic labelling process.

9. Retrieval Results

Measure	Result
Average Top-1 TF-IDF similarity (500 examples)	0.3995
Median Top-1 similarity	0.3349
Average Top-5 similarity	0.3442
Top-1 directly relevant (human review)	14%
Top-1 useful (relevance 1 or 2)	67%
Top-5 directly relevant (at least one 2)	34%
Top-5 useful (at least one 1 or 2)	81%
Queries with no useful Top-5 result	19%

Headline retrieval result: 81% of the 100 manually reviewed queries had at least one useful historical case in their top five results. Only 34% had at least one directly relevant case.

I describe this as evidence availability, not retrieval accuracy and not resolution rate. The relevance labels were manual: 0 = unrelated/generic overlap, 1 = same domain or potentially useful, 2 = core problem directly matches.

10. Resolution Actions and End-to-End Agent

Component	Result
Resolution-action extractor	93/100 exact = 93%
End-to-end intent accuracy	34.0% (50-example sample)
Mean intent confidence	0.3987
AUTO-HANDLE	20%
ESCALATE	80%

The conservative escalation policy is a deliberate choice: the system prefers to ask for human review when confidence or historical evidence is weak.

Failure Analysis

The main value of the evaluation was finding where the simple approach breaks.

11. Five Main Failure Modes

1. Weak-label generalization. The classifier gets 94.73% on its weak-label holdout but only 41.33% on the independent Golden Set. This suggests it learned the heuristic label boundaries better than the actual human task. Hypothesis: the weak labels are too correlated with their own rules.
2. APP_PROBLEM overprediction. Many messages containing generic app language are pulled toward APP_PROBLEM. Hypothesis: APP_PROBLEM is the largest weak-labelled class and becomes a convenient destination for ambiguous examples.
3. OTHER is not learned well. OTHER is used mainly as an abstention/fallback during weak labelling, so the classifier has little direct training signal for it. On the Golden Set, OTHER examples were often forced into one of the learned intents. Hypothesis: a real abstention class needs manually labelled data.
4. Lexical similarity is not semantic relevance. TF-IDF can match common words while missing the actual support problem. It can also produce a semantically useful result with a relatively low similarity score. Hypothesis: Twitter vocabulary variation and short messages are difficult for bag-of-words retrieval.
5. Context-poor and multi-intent tweets. Messages such as “Same here”, “It was 11.1”, or location-only replies are hard to classify without previous conversation turns. Other messages contain multiple symptoms, such as an update followed by performance and battery problems. Hypothesis: the single-message design throws away important thread context.

12. Examples That Expose the Problem

Observed message	Why it is difficult
“Same here”	Almost no standalone information; meaning depends on the parent conversation.
“It was 11.1”	Likely an OS/version detail, but there is no problem statement.
“Wales, UK”	Could be a support-routing detail rather than a support intent.
“This new update has completely ruined my phone. Constantly freezing. Battery life is worse.”	Contains update, performance and battery signals; one-label classification loses the multi-intent structure.
“My iPhone keeps showing no service”	Connectivity and app/device vocabulary can overlap under lexical classification.

These failures also explain why a conservative escalation layer is useful. A high-confidence lexical match does not necessarily mean the system understands the whole support context.

13. LLM-as-Judge Pilot

I implemented a Gemini-based judge using the same 0/1/2 retrieval relevance rubric. Because the available free API quota was limited, this was only a 10-example pilot: nine calls completed and human-vs-judge agreement was 55.56%. I do not treat this as a statistically reliable benchmark. Human labels remain the ground truth.

What the Numbers Really Mean

This section is intentionally explicit about the difference between an attractive metric and a trustworthy one.

14. The Misleading Headline Number

94.73% weak-label validation	41.33% independent Golden accuracy
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The most misleading headline number would be the 94.73% intent validation accuracy. On its own, it looks like the classifier is nearly production-ready. But the validation labels were generated by the same weak-labeling approach used to create the training labels.

When the classifier is tested against the independent, manually labelled 196-example Golden Set, accuracy falls to 41.33% and Macro F1 to 41.68%. I therefore interpret 94.73% as evidence that the model learned the weak-label process, not evidence of real-world intent accuracy.

15. The Retrieval Headline Also Needs a Caveat

81% useful@5 is not an 81% resolution rate. It means that at least one of the five retrieved historical cases was manually judged useful for 81% of reviewed queries. Only 34% had at least one directly relevant case.

This distinction matters because the system is supposed to ground its response in historical evidence. A useful case can still require human interpretation, and a directly relevant case does not guarantee that the drafted response will solve the customer's issue.

16. Why I Kept the System Simple

I considered the available hardware, zero API budget, and the requirement that the project be explainable within a short take-home setting. TF-IDF, Logistic Regression, deterministic action extraction, and a rule-based escalation layer are not state-of-the-art choices, but they are transparent and reproducible.

The resulting system can be inspected from the customer message through retrieval, evidence, draft, and decision. That made it possible to measure individual components instead of hiding all behavior behind a single model score.

17. Reproducibility

The default evaluation does not require an external LLM API. From the repository root:

```
.\venv\Scripts\python.exe scripts\run_evaluation.py
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The full harness currently completes in approximately 2.5 minutes on the development machine, well below the requested 15-minute reproducibility target. It runs baseline evaluation, retrieval evaluation, human relevance summary, action extraction tests, and the end-to-end agent evaluation.

Next Steps and Conclusion

If I had one more week, I would improve the evidence and labels before adding more model complexity.

18. One-Week Improvement Plan

Priority	Planned improvement	Reason
1	Add several hundred more human-labelled intent examples, especially APP_PROBLEM hard negatives, OTHER, ambiguous and multi-intent cases.	The Golden Set shows the main weakness is intent generalization.
2	Use embedding retrieval plus a lightweight reranker.	Improve semantic matching beyond lexical overlap.
3	Reconstruct conversation threads and include previous turns.	Short Twitter messages often have insufficient standalone context.
4	Calibrate intent confidence on a separate labelled set.	Make AUTO-HANDLE thresholds correspond better to real error risk.
5	Tie each generated response action to its supporting historical cases and preserve useful support resources where appropriate.	Make response grounding easier to audit.

19. Final Takeaway

The prototype demonstrates the complete workflow requested: classify → retrieve historical evidence → identify supported resolution actions → draft a response → decide whether to auto-handle or escalate.

The strongest result is not a high classifier score. It is that the system can retrieve at least one useful historical example for 81% of manually reviewed queries while also showing that only 34% have a directly relevant example. That gap points clearly to the next engineering problem: improving semantic retrieval and using conversation context.

The intent results similarly show why independent human evaluation matters. The 94.73% weak-label number looks strong, but the 41.33% Golden Set result exposes the actual limitation of the current labelling strategy.

For this prototype, I chose to make these weaknesses visible rather than tune the system until the headline numbers looked better. The next iteration should improve the data and evidence quality first, then increase model sophistication.

20. Submission Notes

Repository documentation includes a reproducible README and a separate decision log. The raw Twitter dataset is not included in the repository; the README specifies the expected location. The core evaluation pipeline is API-free. The Gemini judge is retained only as a small secondary pilot because free-tier quota limited the number of calls.

Report scope: This report summarizes the prototype and its measured limitations. It does not claim production readiness or customer-resolution accuracy.